

LESSON 6

Puzzles & Review

Name: _____ Date: _____

Knights and knaves

On logic puzzles island:

- **Knights** always tell the truth.
- **Knaves** always lie.

Every person is one or the other. Your job: figure out who is who from what they say.

WORKED EXAMPLE

A and B meet you. A says: "Both of us are knaves."

Could A be a knight? If A were a knight, the statement would have to be true — but then both would be knaves, impossible. So A is a **knave**.

Then A's statement is false, so it is *not* true that both are knaves — at least one is a knight. A is a knave, so **B must be a knight**.

Course review checklist

Before the final practice, can you explain each item in your own words?

- Statement vs open sentence vs not-a-statement
- $\neg$, $\wedge$, $\vee$ and inclusive OR
- Building truth tables; tautology vs contradiction
- $p \rightarrow q$; converse, inverse, contrapositive; $\leftrightarrow$
- Valid vs sound; modus ponens & tollens; common fallacies
- De Morgan's laws

Teaching tip: Discuss puzzles out loud. Logic is a language — saying "assume A is a knight and see what breaks" builds the same habit as proof by cases in math.

Practice — Lesson 6

6.1 WARM-UP

Quick review — answer without looking back (then check Lessons 1–5):

1. What is a statement?
2. When is $p \wedge q$ true?
3. When is $p \rightarrow q$ false?
4. Which form always matches the original conditional: converse or contrapositive?

SHORT ANSWERS

6.2 WARM-UP

One person C says: “I am a knave.” Can C be a knight? Can C be a knave? Explain.

KNIGHT OR KNAVE? WHY?

6.3 STANDARD

A says: “B is a knave.” B says: “A and I are different types.” Determine A and B (knight/knave for each). Show your case analysis.

CASE 1 / CASE 2 REASONING

6.4 STANDARD

Mixed skills:

1. Let $p = \mathbf{T}$, $q = \mathbf{F}$. Evaluate $(\neg p \vee q) \rightarrow p$.
2. Write the contrapositive of: “If a quadrilateral is a square, then it is a rectangle.”
3. Is this valid? “If you live in Sydney, you live in Australia. You live in Australia. So you live in Sydney.”

THREE PARTS

6.5 STANDARD

Three friends make claims about a hidden card (suit is hearts or spades; rank is high or low):

- Ada: “The card is hearts OR high (or both).”
- Ben: “If the card is spades, then it is low.”
- Cara: “The card is not both spades and high.”

The card is **spades and low**. Mark each person’s statement true or false. Did anyone use a true conditional with a false hypothesis?

T/F FOR EACH + NOTE ON BEN’S STATEMENT

6.6 CHALLENGE

Three people — X, Y, Z — each either knight or knave. X says: “Y is a knave.” Y says: “Z is a knave.” Z says: “X and Y are both knaves.” Find the only consistent assignment. (Try assuming X is a knight and follow the chain.)

FULL SOLUTION WITH CASES

6.7 CHALLENGE

Real world + synthesis: Find a short quote from news, an ad, or social media. (1) Identify premises and conclusion if it is an argument. (2) Name any fallacy or invalid pattern. (3) Rewrite using at least two of: $\neg$, $\wedge$, $\vee$, $\rightarrow$. (4) Suggest a clearer honest version.

FOUR-PART ANALYSIS

6.8 CHALLENGE

Mini project: Invent a 4-row truth table for an expression you build from $p, q, \neg, \wedge, \vee, \rightarrow$ that matches a real rule (e.g. “free delivery applies when...”). Explain each row in plain English.

EXPRESSION, TABLE, ENGLISH STORY

CONGRATULATIONS

You finished the six-lesson logic course. Revisit any lesson where warm-ups felt easy but challenges felt hard — that gap is where the learning happened.